



Epilepsy

Epilepsy is a chronic, non-communicable brain disorder that can occur at any age. The main characteristic of the disorder is recurrent seizures resulting from abnormal electrical activity in the nerve cells of the brain. During these seizures, a person may temporarily lose consciousness or awareness and may experience involuntary movements, muscle contractions, muscle twitching, changes in sensation, or certain cognitive and psychological changes. The characteristics of seizures vary depending on where they begin in the brain and how far they spread. Therefore, epilepsy does not manifest in the same way in every patient. In addition, experiencing a single seizure does not mean that a person has epilepsy. Epilepsy is defined by recurrent, unprovoked seizures.

Epilepsy is a common disorder and affects approximately 50 million people worldwide. Although it can occur at any age, it is more common particularly during childhood and older age. In children, seizures may not always appear as obvious convulsions. For example, staring blankly for a few seconds, sudden jerking movements, or certain repetitive movements may be signs of a seizure. Therefore, observing how a seizure occurs, particularly in children, is important for diagnosis.

Epilepsy does not have a single cause. While the cause can be identified in some patients, the cause of the disorder remains unknown in approximately half of cases worldwide. Brain developmental abnormalities, brain damage occurring before or during birth, severe head trauma, brain tumors, brain infections, stroke, certain genetic conditions, and processes associated with the immune system are among the causes associated with epilepsy. Genetic factors also play an important role in epilepsy; however, genetic predisposition does not apply to every patient.

In recent years, significant advances have been made in understanding the genetic aspects of epilepsy. Research has shown that some epilepsies are associated with changes in specific genes. In particular, it has been demonstrated that certain genetic changes occurring during early brain development may play a role in brain developmental disorders that can lead to treatment-resistant epilepsy. In one study, 69 mutated genes associated with such brain developmental disorders were identified, 60 of which had not previously been associated with these disorders. These findings indicate that the genetic structure of epilepsy is highly complex and that many genetic factors that have not yet been discovered may exist.

Advances in genetic research contribute not only to understanding the causes of the disorder but also to its diagnosis and treatment. Genetic testing has now become an important consideration for individuals with unexplained epilepsy. Identifying a genetic cause may provide

information about the course of the disorder and, in some cases, may influence the choice of treatment. However, it is also stated that a specific and definitive treatment is not yet available for every genetic epilepsy.

Observations regarding how a seizure occurs are highly valuable in the diagnosis of epilepsy. When considered necessary by the physician, EEG, brain imaging methods, and blood analyses may be used. In particular, accurately reporting information such as the duration and timing of the seizure, the nature of the movements, and what the person experienced during the seizure can assist in diagnosis.

Epilepsy is treatable, and seizures can be controlled in most patients. According to the World Health Organization, up to 70% of people with epilepsy can become seizure-free with appropriate antiepileptic drug treatment. However, the success of treatment varies depending on the cause of the disorder and the individual patient. In addition to drug treatment, surgical methods and certain device-based treatments may be used in appropriate patients. Surgical intervention may be beneficial in some patients who do not respond adequately to medication.

Epilepsy is not limited to seizures; it can also affect patients' daily and social lives. In addition to physical problems such as falls, injuries, and similar complications

associated with seizures, psychological problems such as anxiety and depression may also occur. Furthermore, due to misconceptions and prejudices in society, people with epilepsy may face discrimination in education, employment, and social life. Therefore, having accurate information about epilepsy is important in preventing patients from becoming socially isolated and facing unnecessary prejudice.

In summary, epilepsy is a disorder caused by abnormal electrical activity in the brain that leads to recurrent seizures and may present with different symptoms in different individuals. Its causes may include various genetic and environmental factors, while the cause remains unexplained in some cases. Significant advances have been made in the diagnosis and treatment of epilepsy, and genetic research has particularly contributed to a better understanding of its causes and different types. With appropriate diagnosis and treatment, many people with epilepsy can control their seizures and continue their lives.

References:

- 1) World Health Organization. (2024, February 7). *Epilepsy*.
<https://www.who.int/news-room/fact-sheets/detail/epilepsy>
- 2) National Institute of Neurological Disorders and Stroke. (2023, April 10). *Researchers uncover secrets of genetic mutations contributing to disorders of the brain*. National Institutes of Health.
<https://www.nimh.nih.gov/news/science-updates/2023/researchers-unlock-genetic-mutations-contributing-to-disorders-in-the-brain>
- 3) Yeditepe University Hospitals. (2026, April 14). *Epilepsy: What is epilepsy?*
<https://yeditepehastaneleri.com/en/health-guide/diseases-treatments/what-epilepsy>
- 4) Ruggiero SM, Xian J, Helbig I. The current landscape of epilepsy genetics: where are we, and where are we going? *Curr Opin Neurol*. 2023 Apr 1;36(2):86-94. doi: 10.1097/WCO.0000000000001141. Epub 2023 Feb 10. PMID: 36762645; PMCID: PMC10088099.

Ali Bocakkıtay/ Founder, Wionpav Labs/ Medical Student, Faculty of Medicine, Selçuk University